QP CODE: **Reg.No: ……..**

Name : ……

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular Examination October 2026

Third Semester

Skill Development Components (SDC)- MG3SDCSDS200 – SOFTWARE ENGINEERING &

SOFTWARE TESTING PRINCIPLES

(2025 ADMISSION ONWARDS)

Duration :2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A),Analyse (An),Evaluate (E),Create(C)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1.	Define Software Engineering and state its main objective?	[K][1]
2.	A software project has fixed and clearly defined requirements. Which development model would you choose? Give one reason?	[U][1]
3.	Mention any two size estimation techniques?	[K][1]
4.	What is code review? List any two defects that can be identified during code review.	[K][2]
5.	Define verification and validation?	[U][2]
6.	What is a Test Case? Mention any two components of a test case?	[U][2]
7.	List the four levels of software testing?	[K][3]

8.	Distinguish between Alpha testing and Beta testing?	[U][3]
9.	State objective of statement coverage in white box testing?	[U][3]
10.	Expand CMMI and state it's meaning?	[K][4]
11.	What is Regression Testing? State its purpose	[K][4]
12.	Mention any two benefits of software reengineering?	[U][4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

13.	Explain Function Point Analysis (FPA). Describe the five functional components used in FPA with suitable examples?	[U][1]
14.	Define module coupling and module cohesion. Explain their types with suitable examples?	[U][1]
15.	Describe the phases of STLC with a suitable example of testing a login system?	[A][2]
16.	A login module contains separate functions for validating username and password. How would you apply Unit Testing to verify these functions? Describe the steps of Unit Testing and design suitable test case?	[U][3]
17.	A web application needs to be tested across multiple browsers and platforms. Explain how Selenium WebDriver can be used for automation testing. Discuss its major features and applications of Selenium with suitable examples?	[An][4]
18.	Explain the different types of software maintenance with suitable examples?	[U][4]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

19.	What is Software Development Life Cycle (SDLC)? Describe the different phases of SDLC and mention the main activities in each phase?	[U][1]

20. Explain the concepts of Static Testing and Dynamic Testing. Discuss the major techniques of each testing approach with suitable examples? [An][2]

21. Discuss Black Box Testing with reference to Boundary Value Analysis and Equivalence Partitioning, using suitable examples? [An][3]

22. Discuss Software Quality Assurance (SQA) with reference to its activities and goals. Describe SQA standards such as ISO 9001 and CMMI, and discuss the role of reviews and audits in ensuring software quality? [An][4]

[15x2 = 30]